



ABC Analytics Pvt Ltd

Data Driven Decision with AI

AI Data Analysis Report

Generated On	04 August 2026 10:35
Rows	84,550
Columns	61
Duplicate Rows	-
Version	1.0

Prepared by

ABC Analytics Pvt Ltd



Confidential



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Dataset Analysis Report

Executive Summary

This analytical report provides a comprehensive evaluation of a cybersecurity network traffic dataset from a Palo Alto PA-440 firewall device protecting the organization "Demo-IR-Org". The dataset contains 84,550 traffic log records captured over a brief 20-minute window on June 6, 2025, between 11:50:10 and 12:10:00.

The primary objective of this analysis is to assess network traffic patterns, identify potential security risks, evaluate firewall rule enforcement, and uncover operational inefficiencies. A key finding is a high level of data redundancy, with exactly 50% of the dataset consisting of duplicate records, alongside critical indicators of suspicious Command and Control (C2C) threat activity.

Dataset Overview

The dataset consists of firewall traffic logs with the following structural characteristics:

1. Total Rows: 84,550
2. Total Columns: 61
3. Duplicate Rows: 42,225 (50.0% of the dataset)
4. Memory Usage: 170.37 MB
5. Data Types: 19 numerical columns, 36 categorical columns, and 5 datetime columns.

Missing Values Analysis

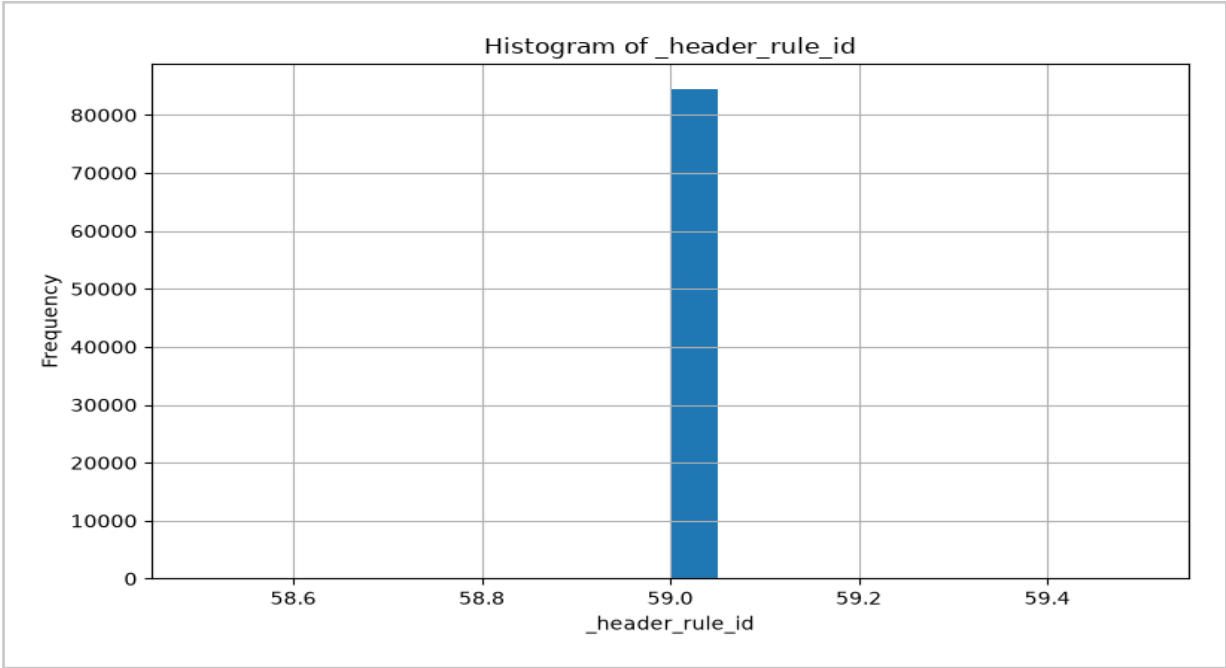


Figure 1. Missing Values Analysis

While most columns are fully populated, missing values are concentrated in threat-related fields and outbound interface logs:

Column Name	Missing Count	Missing Percentage	Description / Impact
`_threat_category`	84,516	99.96%	Populated only during threat events (34 records)
`_threat_family_name`	84,516	99.96%	Populated only during threat events (34 records)
`_threat_ioc`	84,516	99.96%	Populated only during threat events (34 records)
`_threat_malware_ip`	84,516	99.96%	Populated only during threat events (34 records)
`_threat_source`	84,516	99.96%	Populated only during threat events (34 records)



Column Name	Missing Count	Missing Percentage	Description / Impact
`_threat_type`	84,516	99.96%	Populated only during threat events (34 records)
`_threat_victim_ip`	84,516	99.96%	Populated only during threat events (34 records)
`outbound_if`	4,990	5.90%	Missing primarily on blocked/denied traffic

Statistical Findings

Key Traffic Metrics

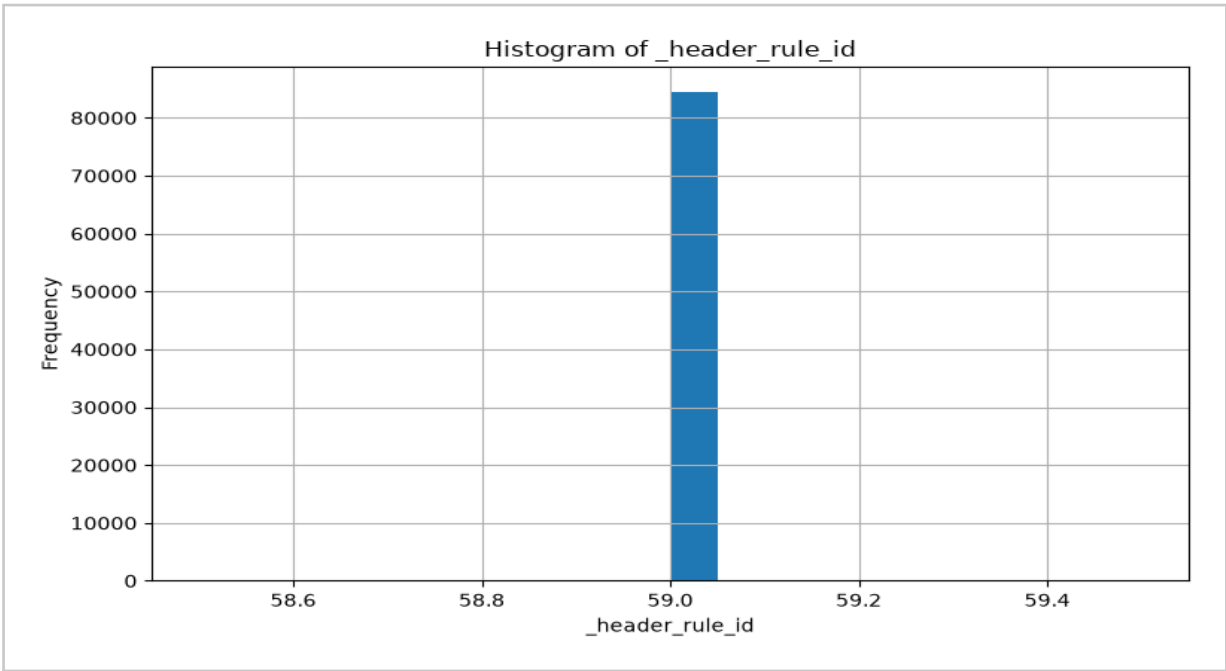


Figure 2. Key Traffic Metrics

The table below summarizes the key numerical metrics of the network traffic sessions:



Metric	Mean	Minimum	50% (Median)	Maximum	Standard Deviation
Bytes Received	30,977.90	0.0	74.0	104,054,808.0	1,088,820.12
Bytes Sent	5,712.05	60.0	106.0	54,619,888.0	357,447.56
Total Bytes	36,689.95	60.0	285.0	104,365,342.0	1,184,440.31
Packets Received	31.18	0.0	1.0	151,414.0	1,069.73
Packets Sent	17.42	1.0	1.0	208,181.0	1,048.14
Elapsed Time (sec)	14.32	0.0	0.0	8,455.0	135.98
Risk Level	2.54	1.0	3.0	5.0	1.10

Correlation Analysis

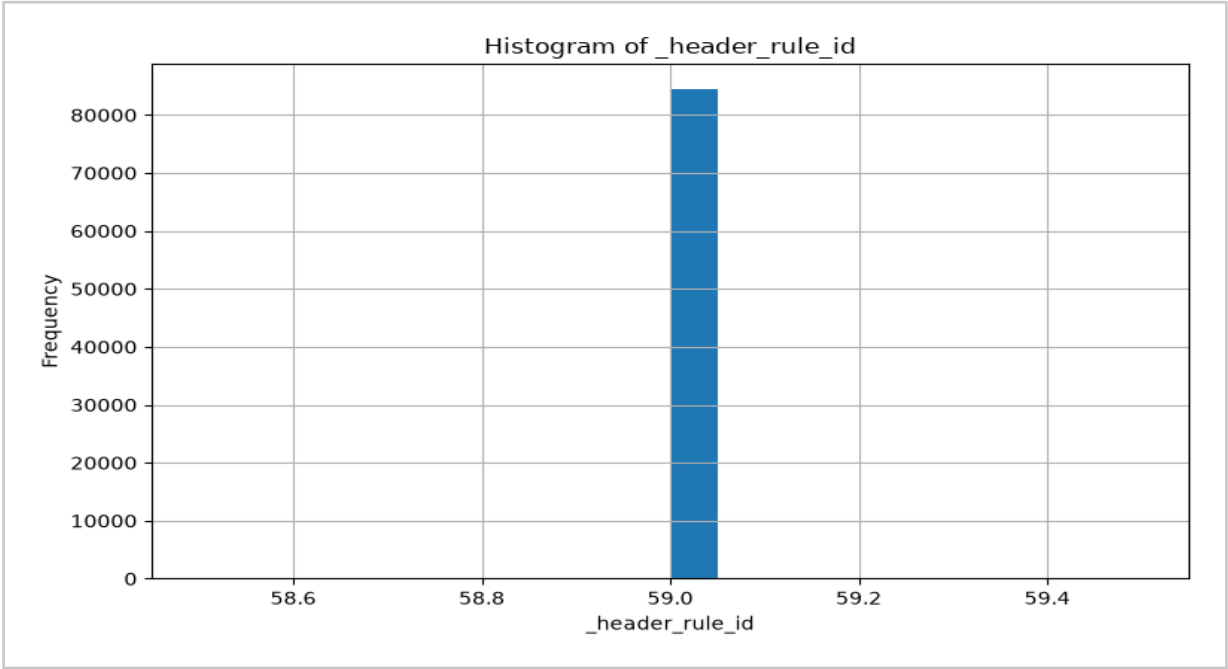


Figure 3. Correlation Analysis

Strong linear relationships exist among network volume metrics, while port configurations show moderate to weak correlations:

1. ``bytes_received`` and ``total_bytes`` (0.954): Extremely strong positive correlation, indicating that received data dominates total session volume.
2. ``pkts_received`` and ``total_bytes`` (0.871): Strong positive correlation between incoming packet counts and total session size.
3. ``bytes_sent`` and ``pkts_sent`` (0.865): Strong positive correlation between outbound bytes and outbound packets.
4. ``bytes_received`` and ``pkts_received`` (0.833): Strong positive correlation between inbound bytes and inbound packets.
5. ``destport`` and ``nat_dest_port`` (0.627): Moderate positive correlation, reflecting standard destination port mapping.
6. ``srcport`` and ``destport`` (-0.334): Weak negative correlation.
7. ``risk_level`` and ``nat_dest_port`` (-0.256): Weak negative correlation, suggesting lower risk levels are associated with higher-numbered NAT destination ports.

Outlier Detection

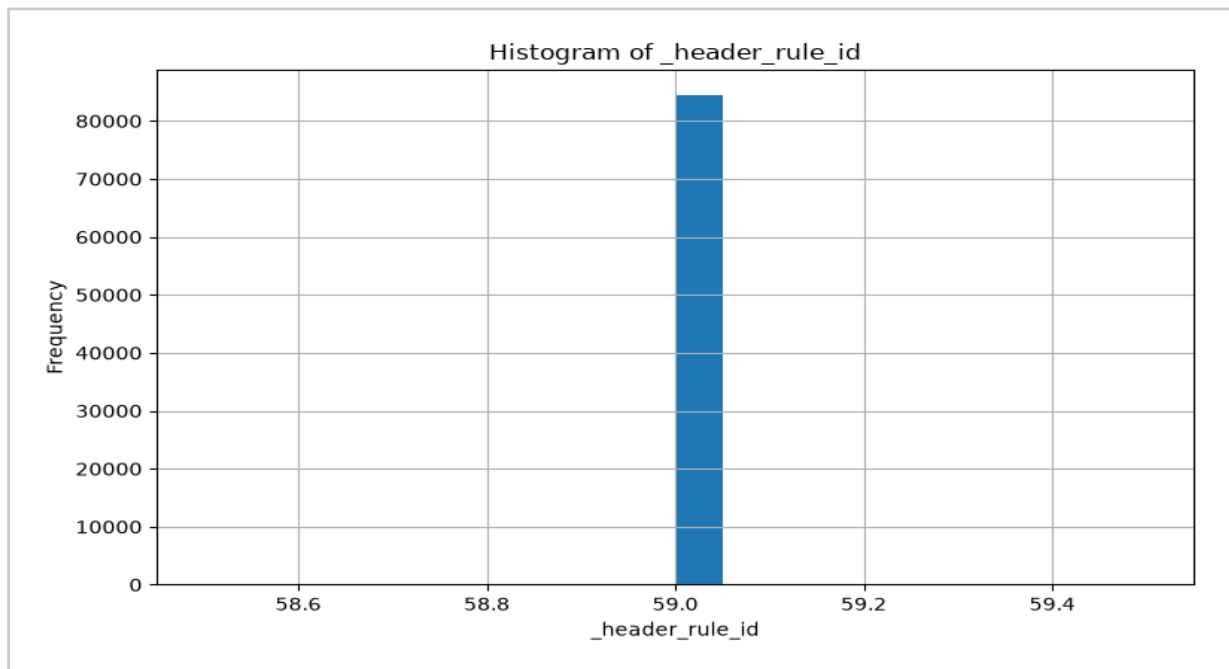


Figure 4. Outlier Detection

Significant outliers exist across several operational metrics, indicating highly asymmetric traffic patterns or potential anomalies:

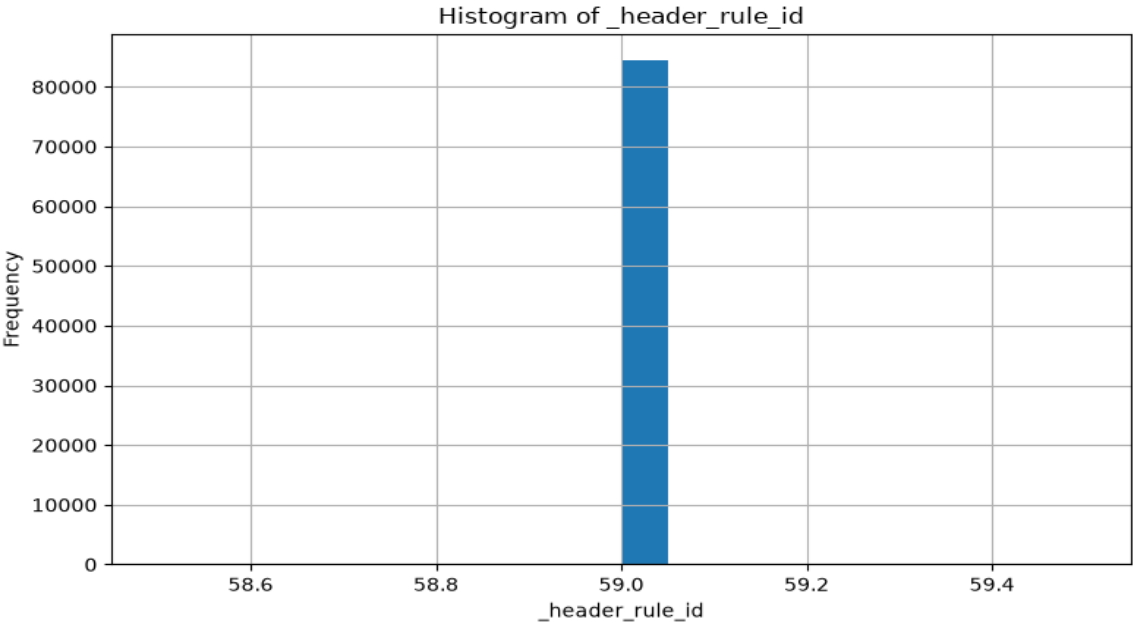
8. Packets Received: 18,334 outliers
9. Elapsed Time: 17,446 outliers
10. Bytes Received: 15,735 outliers



- 11. Packets Sent: 15,619 outliers
- 12. Total Bytes: 12,291 outliers
- 13. Session ID: 10,050 outliers
- 14. Destination Port: 7,432 outliers
- 15. NAT Destination Port: 6,780 outliers
- 16. Source Port: 1,098 outliers

Visual Analysis

histogram header rule id



Header Rule ID Distribution

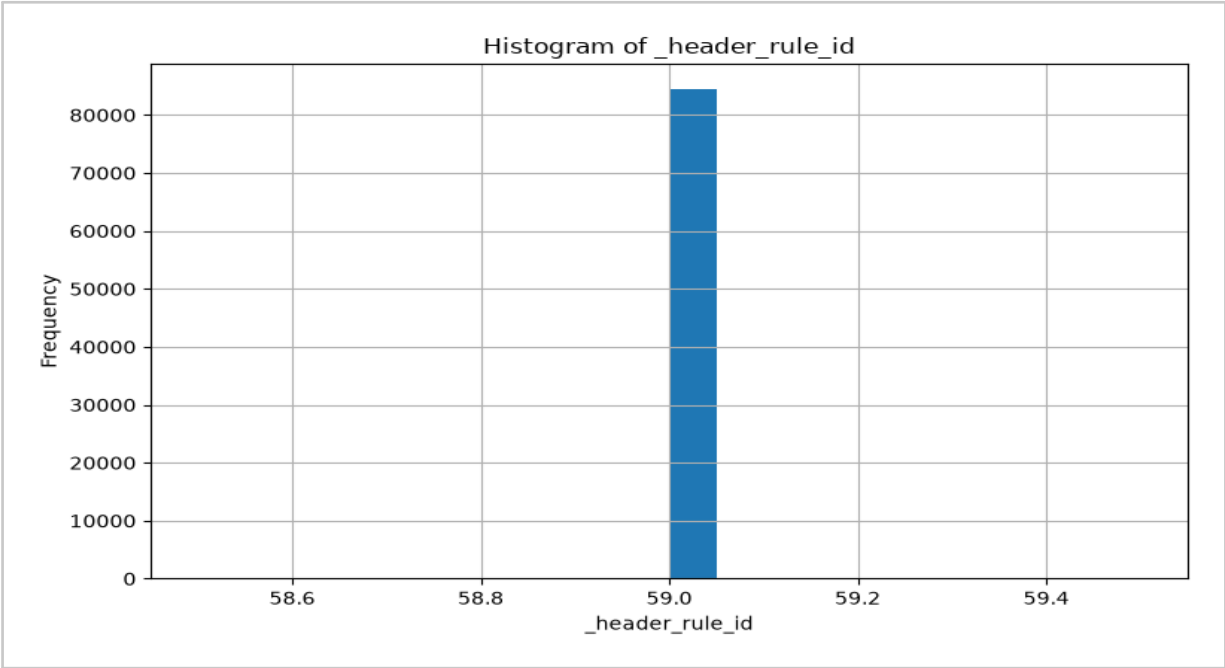


Figure 5. Header Rule ID Distribution

- 1. Visual Asset: `outputs\charts\\_header\_rule\_id\_histogram.png`
- 2. Description: This histogram visualizes the distribution of the `\_header\_rule\_id` field across the dataset.
- 3. Analysis: The statistical summary shows that `\_header\_rule\_id` has a mean, minimum, and maximum value of exactly 59.0 with a standard deviation of 0.0. Consequently, the histogram displays a single, highly concentrated vertical bar at the value of 59.0 for all 84,550 records. This confirms that a single header rule was uniformly applied to every logged traffic event during this 20-minute window.

Business Insights

1. Security Risk Exposure (C2C Threat Activity)

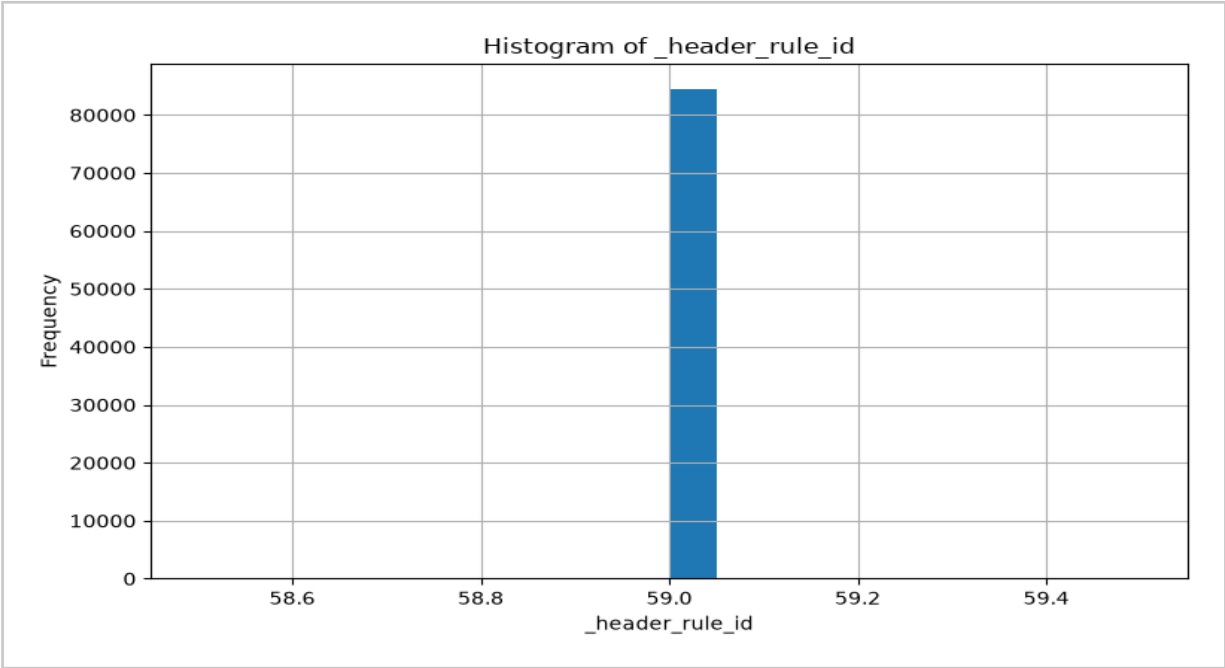


Figure 6. 1. Security Risk Exposure (C2C Threat Activity)

Although threat-related fields are 99.96% empty, the remaining 0.04% represents a critical security event. There are 34 recorded instances of a `[Suspicious]` threat category. These events are classified under the `[Suspicious\_C2C]` (Command and Control) threat family, originating from the user-defined IPv4 malware IP 103.10.24.44 and targeting the internal victim IP 192.168.28.59. This indicates an active, targeted security threat within the network.

2. Permissive Policy Configuration

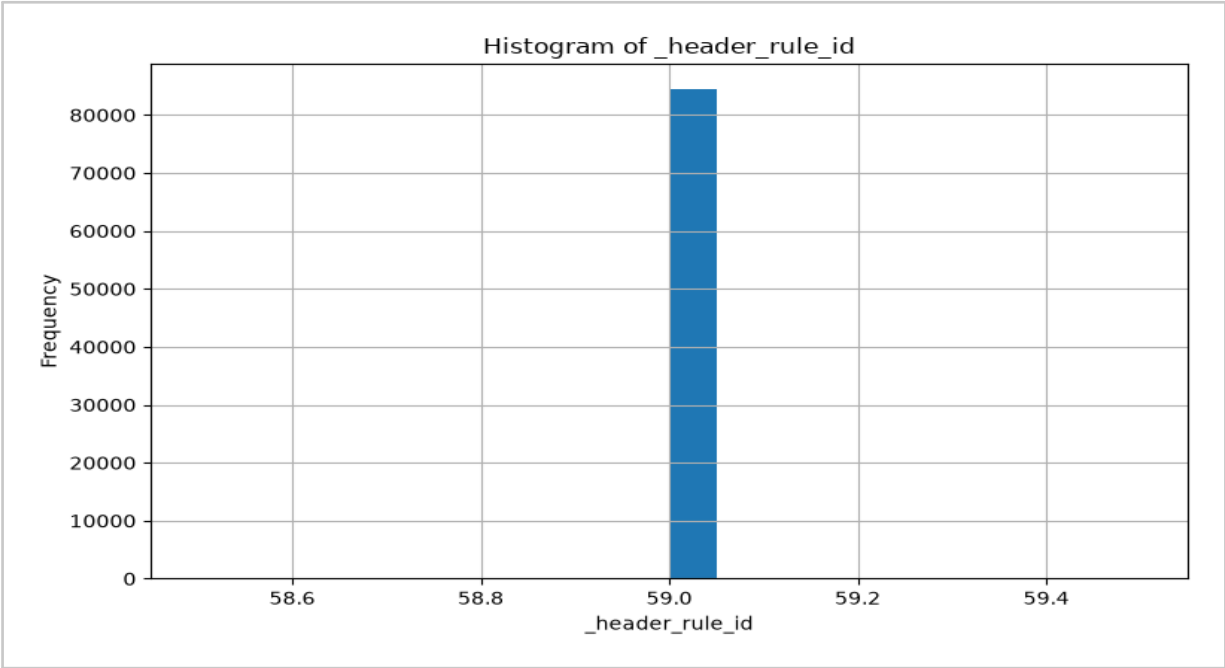


Figure 7. 2. Permissive Policy Configuration

The firewall rule "Allow-any" was triggered 63,321 times (representing approximately 75% of all logs), followed by "Allow-Web-Browsing" (16,197 times). The heavy reliance on a broad "Allow-any" rule suggests an overly permissive security posture that may be exposing the internal network to unnecessary external risks.

3. Traffic Actions and Protocol Dominance

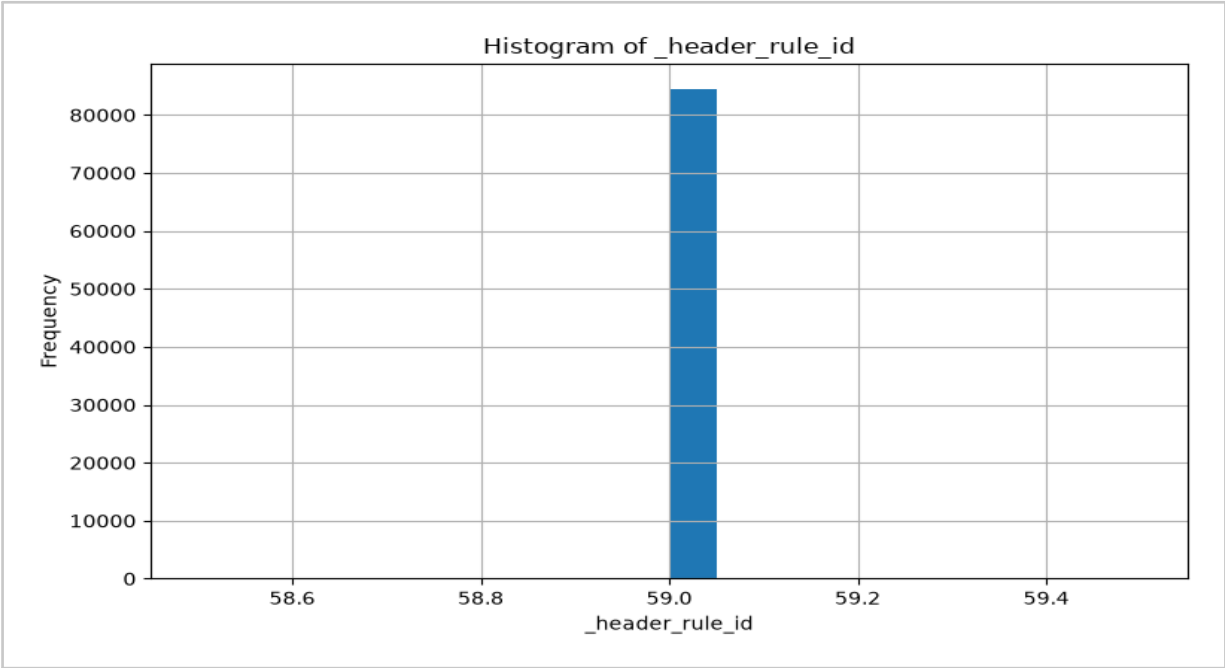


Figure 8. 3. Traffic Actions and Protocol Dominance

1. Actions: The firewall allowed the vast majority of traffic (79,518 "allow" actions), while blocking a smaller portion (4,990 "deny" actions and 42 "reset-both" actions).
2. Protocols: UDP is the dominant protocol with 52,420 occurrences, followed by TCP (31,408) and ICMP (722).
3. Applications: "dns-base" is the most frequent application (38,076 occurrences), followed by "ssl" (12,202) and "quic-base" (8,945).

4. Geographic and Network Sources

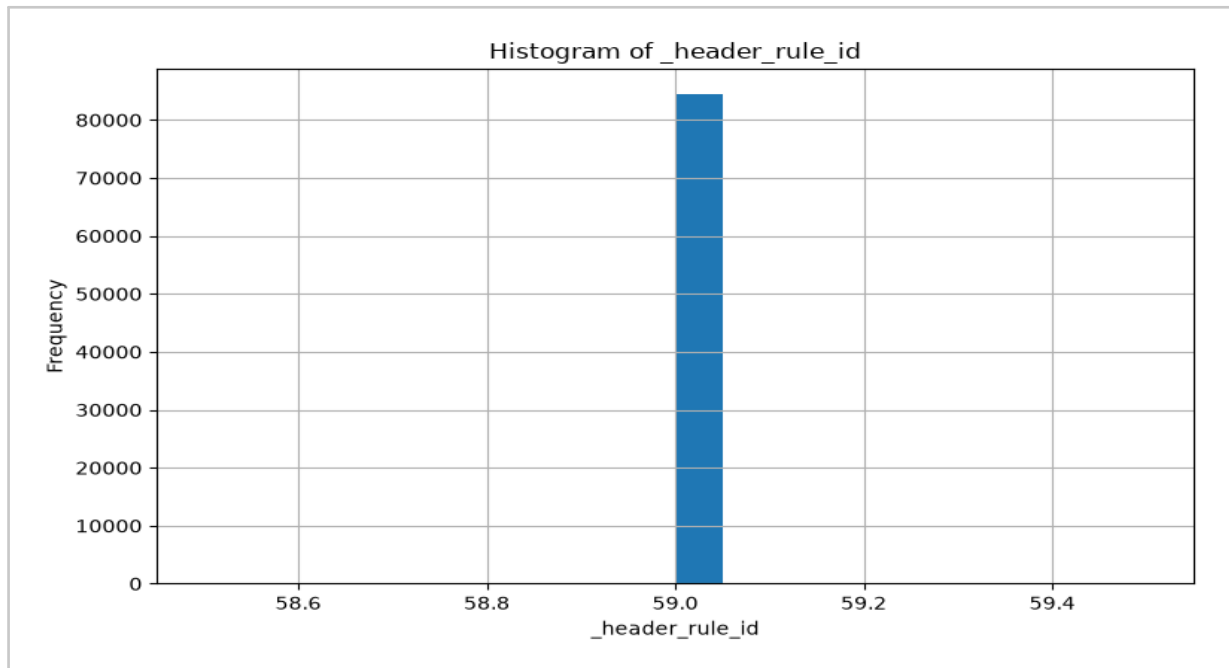


Figure 9. 4. Geographic and Network Sources

The United States is the top source country with 61,175 occurrences, followed by India (11,218) and Australia (8,051). The primary internal source zone is "Trust" (84,550 occurrences), and the top internal source IP is 192.168.29.248 (7,817 occurrences).

5. Data Storage and Processing Overhead

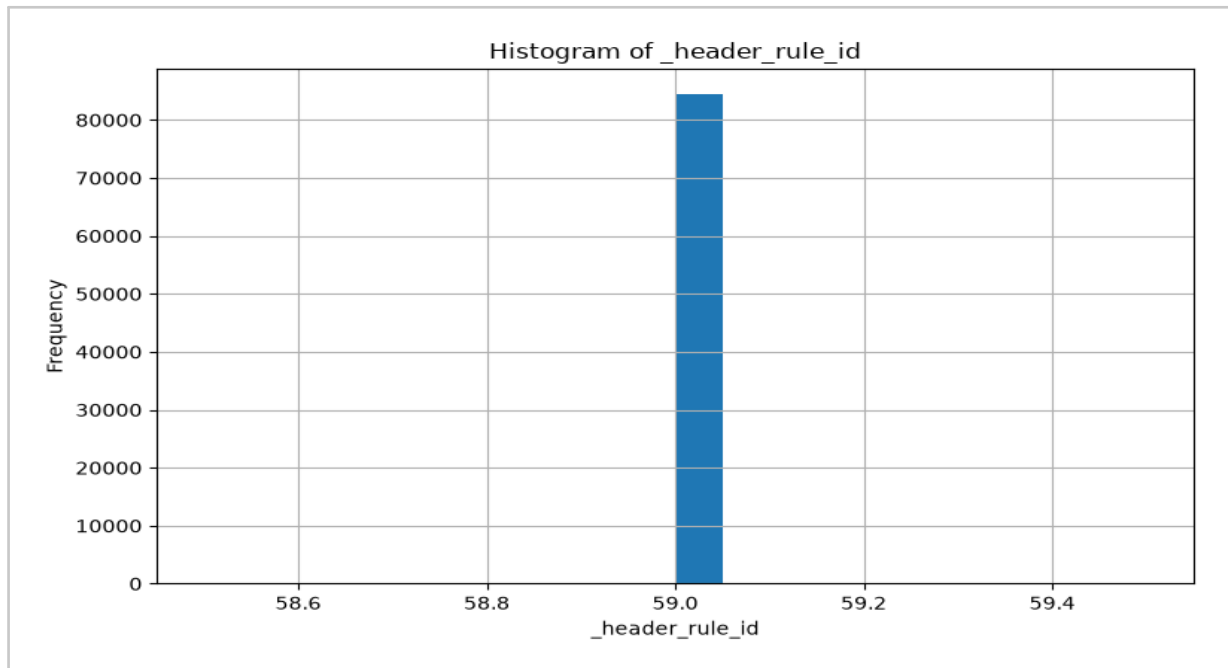


Figure 10. 5. Data Storage and Processing Overhead

Exactly 50% of the dataset (42,225 rows) consists of duplicate rows. This high duplication rate indicates redundant logging or data collection issues, which unnecessarily inflates log storage costs and slows down SIEM (Security Information and Event Management) analysis.

Recommendations

- 1. Isolate and Investigate Victim Host:** Immediately isolate the internal host at IP 192.168.28.59 and investigate the 34 suspicious Command and Control (C2C) threat events associated with the external malware IP 103.10.24.44 to determine if a compromise has occurred.
- 2. Audit and Restrict Firewall Rules:** Review and restrict the "Allow-any" firewall rule (triggered 63,321 times). Transition toward a "least privilege" model by replacing broad rules with specific, application-aware rules.
- 3. Resolve Log Duplication:** Investigate the root cause of the 50% log duplication rate (42,225 duplicate rows) within the logging infrastructure or syslog forwarder to optimize storage capacity and improve processing efficiency.
- 4. Investigate High-Volume Outliers:** Analyze the sessions associated with the extreme outliers in `bytes\_received` (up to 104 MB in a single session) and `elapsed\_time` (up to 8,455 seconds) to ensure they represent legitimate business traffic and not data exfiltration or unauthorized



tunneling.

Conclusion

The analysis of the Palo Alto PA-440 firewall logs for "Demo-IR-Org" reveals a highly active network environment dominated by UDP-based DNS and SSL traffic. While the firewall successfully blocked 4,990 unauthorized connections, the presence of 34 suspicious Command and Control (C2C) events targeting an internal host represents an immediate security risk. Furthermore, the high volume of triggers on the "Allow-any" rule and a 50% log duplication rate highlight critical areas for policy tightening and log infrastructure optimization. Implementing the recommended security investigations and rule audits will significantly strengthen the organization's security posture.